Revisiting Isolated DC-DC converters-I

Introduction

The rapid adoption of Electric Vehicles (EVs) necessitates robust and efficient charging infrastructure. At the heart of every Electric Vehicle Supply Equipment (EVSE) lies a sophisticated power conversion unit, responsible for transforming grid power into a suitable form for EV batteries. This unit typically comprises an AC-DC converter followed by an isolated DC-DC converter. While the AC-DC stage handles initial power factor correction and voltage regulation, the isolated DC-DC converter plays a crucial role in ensuring both precise control over the charging process and, critically, galvanic isolation for user safety.

This lecture delves into the fundamental principles and common topologies of isolated DC-DC converters, focusing on their operational characteristics and design considerations within the context of EV charging. Understanding these converters is paramount for developing high-performance, compact, and safe charging solutions that meet the demanding requirements of modern EVs. We will revisit the forward converter, analyzing its operation, advantages, and inherent limitations, setting the stage for exploring more advanced isolated topologies.

Core Concepts

Term	Definition	Significance
`Isolated DC-DC Converter`	An electronic circuit that converts a DC voltage from one level to another while providing electrical isolation between the input and output.	Essential for safety (galvanic isolation) and precise control of battery charging parameters in EVSEs.
`Galvanic Isolation`	A principle of isolating functional sections of electrical systems to prevent current flow between them, often achieved using transformers.	Provides critical safety by protecting users from potential electrical hazards from the grid during EV charging.

Term	Definition	Significance
`High-Frequency Transformer`	A transformer designed to operate efficiently at switching frequencies much higher than the utility line frequency (e50/60 Hz).	Significantly reduces the size and weight of the transformer, enabling compact and power-dense isolated DC-DC converters.
`Volt-Second Balance`	The principle stating that the average voltage across an inductor over one switching period must be zero in steady-state operation.	Fundamental for deriving the input-output voltage relationship of DC-DC converters and ensuring stable operation without core saturation.
`Core Saturation`	A condition where a magnetic core can no longer support an increase in magnetic flux density, leading to a sharp drop in inductance.	Must be avoided in transformers and inductors as it causes high currents, increased losses, and potential component damage.
`B-H Curve`	A graph showing the relationship between magnetic flux density (B) and magnetic field strength (H) for a magnetic material.	Used to characterize magnetic materials and understand their behavior, including saturation limits and core utilization in transformers.
`Forward Converter`	An isolated DC-DC converter topology that transfers energy from input to output during the switch's ON time, utilizing a transformer.	A basic isolated topology suitable for low-to-medium power applications, but with specific limitations regarding core reset and switch stress.

Detailed Analysis

1. The Power Conversion Unit in EVSE

The power conversion unit within an EVSE is a critical component, typically structured in stages to achieve specific objectives. Initially, an AC source (grid) feeds an AC-DC converter. This front-end converter is responsible for converting the alternating current from the grid into a regulated DC voltage. Key objectives for this

stage include:
* **Output Voltage Regulation**: Ensuring the DC output voltage is stable and at a desired level.
* **Unity Power Factor**: Drawing current from the grid in phase with the voltage to minimize reactive power and improve efficiency.
Previous discussions covered various AC-DC converter types, including single-phase and three-phase topologies, boost PFC converters (like totem pole and flyback-based PFC), and their control strategies, such as vector control for three-phase systems.

Following the AC-DC stage, an isolated DC-DC converter is cascaded. This second stage is crucial because most AC-DC converters do not inherently provide galvanic isolation. The isolated DC-DC converter serves two primary functions:
* **Output Voltage and Current Control**: Precisely regulating the voltage and current delivered to the EV battery, which is essential for optimal charging and battery longevity.
* **Galvanic Isolation**: Providing electrical separation between the grid and the battery. This is a critical safety feature, protecting users from potential electric shock hazards.

2. Types of Isolated DC-DC Converters

A variety of isolated DC-DC converter topologies exist, each suited for different power levels and application requirements:
* **Low Power (up to ~500W)**:
* **Forward Converter**: Often used in Switched-Mode Power Supplies (SMPS) for low power levels (e.g., 100-200W).
* **Flyback Converter**: Also suitable for low power applications, sometimes used for PFC.
* **Push-Pull Converter**: Can handle slightly higher power levels, typically up to 500W, but generally not preferred beyond that due to operational complexities.
* **Higher Power (for EV charging)**:
* **Half-Bridge Converter**: Generally used for low to medium power applications.
* **Full-Bridge Converter**: Commonly employed for high-power applications, including EV charging, offering unidirectional power flow.
* **Phase-Shifted Full-Bridge (PSFB) Converter**: A variation of the full-bridge converter where switches are modulated differently to achieve soft switching, improving efficiency.
* **Dual Active Bridge (DAB) Converter**: A highly efficient topology capable of bidirectional power flow, making it ideal for EV charging and vehicle-to-grid (V2G) applications. Variations include resonant-based DAB and multi-level DAB for high-voltage operations.

* **Resonant-Based Converters**: Such as LLC or CLLC converters, which utilize resonant tanks to achieve high efficiency and soft switching characteristics.

3. The Imperative for High-Frequency Isolation

A fundamental question arises: why not place a low-frequency transformer (e.g., 50 Hz) directly after the AC source for isolation, followed by a simple non-isolated DC-DC converter? The answer lies in the physics of transformers and their size.
* **Faraday's Law**: The voltage induced in a transformer winding is given by $V = N * d\Phi/dt$, where N is the number of turns and $d\Phi/dt$ is the rate of change of magnetic flux.
* **Flux Density and Frequency**: The change in flux ($\Delta\Phi$) is proportional to the change in flux density (ΔB) and the core's cross-sectional area (A). The rate of change ($d\Phi/dt$) is inversely proportional to the period (T) or directly proportional to the frequency (f). Thus, $V \approx N * \Delta B * A * f$.
* **Core Area and Frequency Relationship**: For a given voltage (V), number of turns (N), and maximum allowable flux density swing (ΔB, limited by core saturation), the core's cross-sectional area (A) is inversely proportional to the operating frequency ($A \propto 1/f$).
* **Implication for Design**: If a transformer operates at the grid frequency (50 Hz), it requires a significantly larger core area compared to one operating at high frequencies (e.g., tens or hundreds of kHz). By converting DC to high-frequency AC before the transformer, the isolated DC-DC converter can utilize a much smaller, lighter, and more compact high-frequency transformer. This leads to a higher power density and reduced overall volume and weight of the power conversion unit, which is crucial for practical EVSE designs.

4. Operation of the Forward Converter

The forward converter is a buck-derived isolated topology. It consists of a transformer with primary (Np) and secondary (Ns) windings (turns ratio n = Ns/Np), a main switch (S1), a reset winding with diode (DS1) and resistor (Rf), two output diodes (D1, D2), an output inductor (L), and an output capacitor (C).

a. DTS Period (Switch S1 ON)
* **Switch State**: S1 is turned ON.
* **Primary Side**: The input voltage (Vin) is applied across the primary winding (Vp = Vin).
* **Secondary Side**: Due to transformer action, a voltage nVin appears across the secondary winding (Vs = nVin). The dot polarity ensures D1 is forward-biased and D2 is reverse-biased.
* **Output Inductor**: The voltage across the output inductor (VL) is nVin - Vo. Since

nVin > Vo (for buck-derived operation), VL is positive, causing the inductor current (IL) to rise linearly.
* **Transformer Flux**: The constant positive voltage across the primary winding causes the magnetic flux (Φ) in the transformer core to rise linearly.
* **Current Path**: Current flows from Vin, through S1, through the primary winding, and then through the secondary winding, D1, L, C, and the load. The current through S1 includes the reflected load current and the magnetizing current (IM) of the transformer.

b. (1-D)TS Period (Switch S1 OFF)
* **Switch State**: S1 is turned OFF.
* **Primary Side**: The primary winding current (including magnetizing current) attempts to flow, but S1 is open. The magnetizing current now flows through the reset winding, DS1, and Rf. This provides a path for the transformer core's flux to reset.
* **Secondary Side**: As the primary voltage collapses, the polarity across the secondary winding reverses. D1 becomes reverse-biased, and D2 becomes forward-biased.
* **Output Inductor**: The output inductor current (IL) now freewheels through D2, the output capacitor, and the load. The voltage across the inductor (VL) is -Vo, causing IL to decrease linearly.
* **Transformer Flux Reset**: The magnetizing current flowing through Rf causes the flux in the transformer core to decay exponentially, ideally returning to zero by the end of the (1-D)TS period.
* **Volt-Second Balance**: For the output inductor, the average voltage over a switching period (Ts) must be zero:
(nVin - Vo) * DTs + (-Vo) * (1-D)Ts = 0
nVinD - VoD - Vo + VoD = 0
nVinD - Vo = 0
Vo = n * Vin * D*
This equation shows that the output voltage is a function of the input voltage, the transformer turns ratio, and the duty cycle (D), similar to a buck converter.

5. Disadvantages of the Forward Converter

Despite its simplicity, the forward converter has several limitations that restrict its use, especially in high-power or high-frequency applications:
* **Risk of Core Saturation**: The transformer core must be completely reset to zero flux during each switching cycle. If the (1-D)TS period is insufficient, or if the reset circuit (DS1, Rf) is improperly designed, the flux will not fully reset. Over several cycles, a DC offset in the flux can build up, eventually leading to core saturation. This is a critical failure mode, causing high magnetizing currents and potential

damage.
* **Limited Switching Frequency**: To ensure complete flux reset, a sufficient duration for the (1-D)TS period must be provided. This requirement limits the maximum achievable switching frequency, as increasing frequency reduces the available time for reset.
* **Diode Reverse Recovery Losses**: The output diodes (D1, D2) experience reverse recovery effects, especially D1 when it turns off. These losses increase significantly with higher switching frequencies, reducing overall efficiency and generating heat.
* **Lossy Reset Mechanism**: The reset circuit typically uses a resistor (Rf) to dissipate the stored magnetizing energy. This results in I^2R losses, making the reset process inherently lossy and reducing efficiency.
* **Limited Core Utilization**: The forward converter utilizes only one quadrant of the transformer's B-H curve (from zero to Bmax). This means that for a given power level, a larger core is required compared to topologies that utilize both positive and negative flux swings (e.g., push-pull, full-bridge), which can swing from -Bmax to +Bmax. This leads to a less efficient use of the magnetic material.
* **High Switch Voltage Rating**: When the main switch S1 turns off, it must block not only the input voltage (Vin) but also the reflected voltage from the reset winding (IM*Rf) and the voltage induced by the decaying magnetizing current. Consequently, the voltage stress across S1 can be significantly higher than Vin, requiring a higher voltage rating for the switch, which can increase cost and conduction losses.

These disadvantages highlight why the forward converter is generally limited to lower power applications and why more advanced isolated topologies are preferred for high-power EV charging systems.

Key Takeaways

- [ ] Isolated DC-DC converters are essential in EVSEs for both output control and critical galvanic isolation.
- [ ] Galvanic isolation provides safety by electrically separating the EV battery from the grid.
- [ ] High-frequency transformers are used in isolated DC-DC converters to significantly reduce the size and weight of the magnetic components.
- [ ] The core area of a transformer is inversely proportional to its operating frequency ($A \propto 1/f$).
- [ ] The forward converter is a buck-derived isolated topology that transfers energy during the switch's ON time.
- [ ] Volt-second balance across the output inductor determines the output voltage ($Vo = n * Vin * D$).

- [ ] The transformer core in a forward converter must be fully reset in each cycle to prevent saturation.
- [ ] Forward converters suffer from limitations such as core saturation risk, limited switching frequency, lossy reset, and high switch voltage stress.
- [ ] Core utilization in a forward converter is limited as it only uses one quadrant of the B-H curve.

Conclusion

The isolated DC-DC converter is an indispensable component in the EV charging ecosystem, bridging the gap between the grid-connected AC-DC stage and the battery. Its dual role of precise power delivery and crucial galvanic isolation underscores its importance for both performance and safety. While the forward converter provides a foundational understanding of isolated power transfer, its inherent limitations, particularly concerning core reset, switching frequency, and component stress, restrict its application to lower power levels.

The need for higher power density, improved efficiency, and bidirectional power flow in modern EV charging systems drives the development and adoption of more sophisticated topologies like the full-bridge, phase-shifted full-bridge, and dual active bridge converters. These advanced designs address the shortcomings of simpler converters, paving the way for more compact, efficient, and versatile EV charging solutions that can meet the evolving demands of the automotive industry. The continuous innovation in isolated DC-DC converter technology is vital for the widespread and safe deployment of electric vehicles.